

MANSWI PRADHAN

B.Tech Mathematics & Computing — IIIT Raichur

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PROFILE

Second-year Mathematics & Computing student with experience in AI, statistical reasoning, cybersecurity and technical analysis. Skilled at breaking down complex problems, evaluating outputs against evidence, identifying errors and clearly documenting reasoning.

EDUCATION

Indian Institute of Information Technology (IIIT), Raichur

Sep 2025 – Sep 2029 (Expected)

B.Tech in Mathematics & Computing · Currently in 2nd year

EXPERIENCE

Sponsorship Team Member — TEDxIIIT Raichur

Jun 2026 – Present

- Research and identify potential sponsors aligned with the event's vision and audience; build a database of prospective partners and key decision-makers.
- Support sponsorship outreach through market research, prospect qualification and relationship building.

Core Member — EHACS (Ethical Hacking & Cybersecurity Community)

Mar 2026 – Present

- Helped organize and run a campus-wide cybersecurity treasure hunt (48 players, 8 teams) with layered cryptographic puzzle mechanics.
- Manage the club's LinkedIn page and support its online outreach and visibility.

Core Member — Mathematics Club, IIIT Raichur

Feb 2026 – Present

- Organized and coordinated Pi Day celebrations featuring interactive mathematics exhibits and student-led projects.
- Manage the club's LinkedIn presence, creating and publishing content highlighting events and initiatives.

PROJECTS

CTF Learning Assistant — Solo Project

- Built a browser-based encoding detector covering 10+ formats (Base64, Hex, Binary, ROT13, Vigenère, Morse, Octal, A1Z26, URL, XOR/Base32) that teaches the underlying concept before revealing the decode.
- Implemented chi-squared frequency analysis to auto-crack Caesar ciphers without brute-forcing all 26 shifts.

MediFind — AI-Powered Medical Chatbot (3-person team, supervised by Dr. Kaushiki Roy)

- Built an AI-powered chatbot that triages symptoms and recommends nearby doctors via specialty-matching.
- Owned the full frontend, a semantic specialist-matching layer using Sentence Transformers and symptom-extraction/classification (Groq API, FastAPI) with a keyword-based fallback.

Fruit Recognition — VGG16 Transfer Learning (3-person team)

- Designed a 15-class fruit image classifier on a transfer-learned VGG16 backbone (frozen conv layers, custom classifier head), comparing Cross-Entropy vs. Focal Loss.
- Built and ran the training loop - best experiment reached 97% test accuracy.

picoCTF Writeups — Solo

- Authored walkthroughs across Forensics, Cryptography, and General Skills categories, documenting tools used and dead ends hit, not just the final solve.

COMPETITIVE PROGRAMMING & CTF

- CodeChef (newton_lebniz): Rookie league, 882 rating (Div 4), 1246 DSA rating, 110+ problems solved across 8 rated contests.
- Badges: Contest Contender (Bronze), Problem Solver (Bronze - 50+ problems).
- Active in picoCTF and campus CTF challenges, focused on cryptography and forensics categories.

TECHNICAL SKILLS

- **Programming:** C, Java, Python, MATLAB
- **Security:** CTFs, cryptography, frequency/statistical analysis, Linux
- **Machine Learning:** PyTorch, Sentence Transformers, transfer learning
- **Web / Backend:** HTML/CSS, API, MySQL
- **Tools:** Git, Linux

ADDITIONAL BACKGROUND

- Entered the program with a Mass Communication background — brings a communication-first approach to technical documentation and writeups.
- **Languages:** English (professional working proficiency), Hindi (full professional proficiency), Marathi (native or bilingual)